06 · Pedigree charts: she says they will be tested

The words of this track, as it says them. Bold brown is the examiner.

Pedigree charts. Five marks.

Explain how you work out the mode of inheritance from a pedigree chart, how you assign genotypes, and how you find the chance of another affected child. Use an autosomal dominant family as your example: an affected father and an unaffected mother, who have both affected and unaffected children.

In her own words, printed on six of her quiz questions: pedigree charts will be tested in Test 3. So this one is close to certain, either written out like this, or as a chart you have to read.

The hook is three questions, from Emma's slides. One: does it skip a generation? Two: is there a difference between the sexes? Three: who are the homozygous recessives? Then a Punnett square.

Step one, read the chart. Squares are males. Circles are females. Shaded means affected. Half shaded, or a dot, means a carrier. A line through a symbol means the person has died. The generations are Roman numerals down the side, and the people are numbered from left to right. And when you check the generations, leave out the people who married in.

Step two, dominant or recessive? If the trait appears in every generation and never skips, it's dominant. If it skips a generation, it's recessive. The giveaway for recessive is an affected child with two unaffected parents.

Step three, autosomal or X-linked? Count the affected males and the affected females. About equal: autosomal. A clear difference: sex-linked. And one quick test: if a father passes it to his son, it can't be X-linked, because a son gets his father's Y, not his X.

The X-linked rules, in case the chart is lopsided. In X-linked recessive, fathers can't pass it to their sons, and an affected mother passes it to all of her sons. In X-linked dominant, an affected father passes it to all of his daughters.

Step four, genotypes. Label the homozygous recessives first: little b, little b. Everyone else has at least one dominant allele, and you work out their second one from their parents and their children. In her example, the affected father must be big B little b, because some of his children are unaffected, little b little b, and each of them got a little b from him. The unaffected mother is little b little b.

Step five, the chance. Draw a Punnett square of the parents. Big B little b, crossed with little b little b. Two boxes are big B little b, and two boxes are little b little b. So each child has a fifty per cent chance of being affected.

The traps. Say how you can tell: skips or not, males versus females, father to son. The "how can you tell" part carries its own marks. Second: father to son rules out X-linked. Third: the chance is per child. Fifty per cent for every pregnancy, not one in every two children.

Once more. How do you read a pedigree, assign the genotypes, and get the chance?

Squares are males, circles are females, shaded is affected. In every generation, never skipping: dominant. Skips a generation: recessive. Males and females about equal: autosomal, and father to son rules out X-linked. Label the homozygous recessives first, then work out everyone else from their parents and children: the affected father is big B little b, and the mother is little b little b. The Punnett square gives two big B little b to two little b little b: a fifty per cent chance for each child.